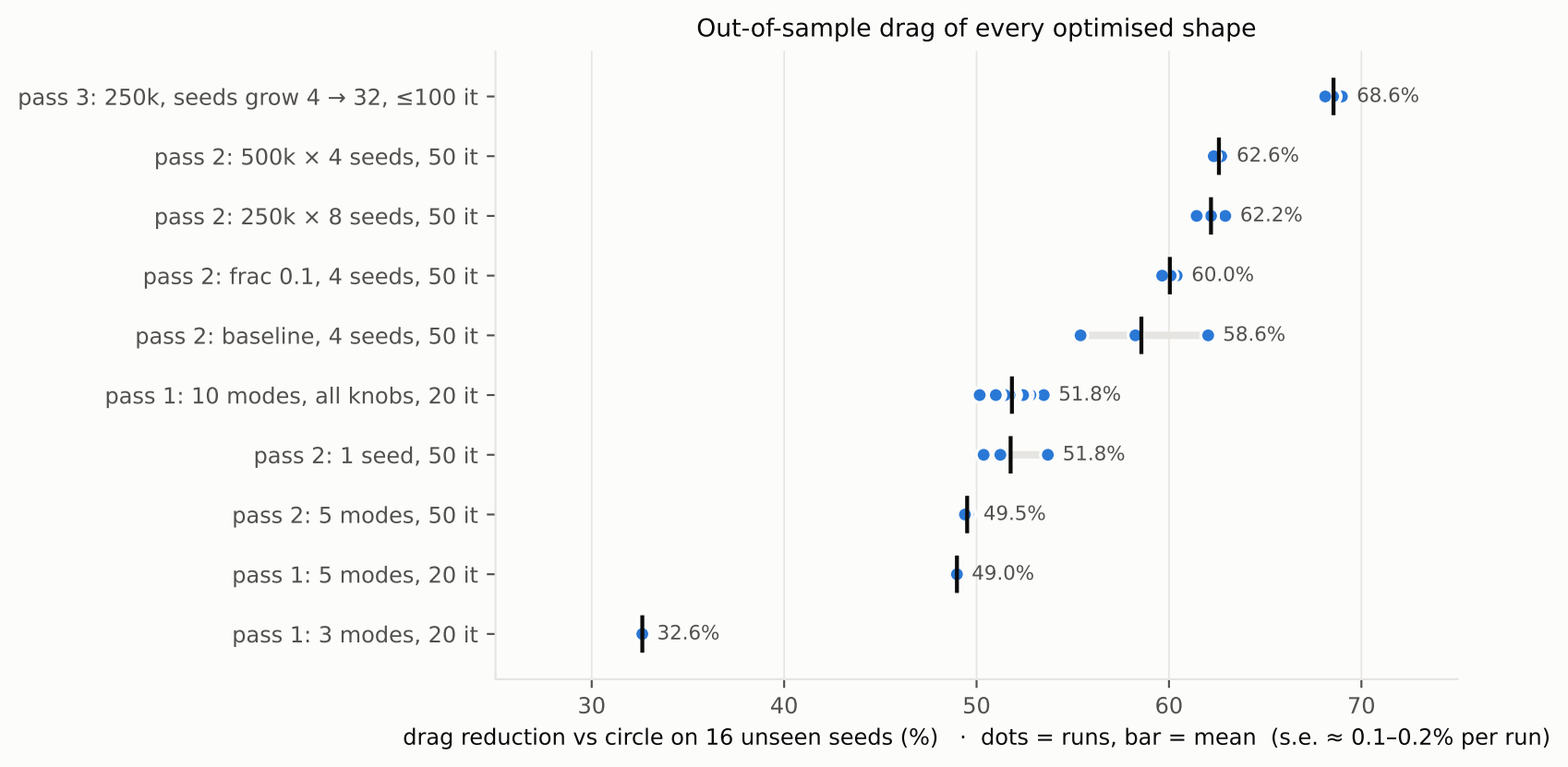
Out-of-sample drag of every optimised shape
pass 3: 250k, seeds grow 4 → 32, ≤100 it
pass 2: 500k × 4 seeds, 50 it
pass 2: 250k × 8 seeds, 50 it
pass 2: frac 0.1, 4 seeds, 50 it
pass 2: baseline, 4 seeds, 50 it
pass 1: 10 modes, all knobs, 20 it
pass 2: 1 seed, 50 it
pass 2: 5 modes, 50 it
pass 1: 5 modes, 20 it
pass 1: 3 modes, 20 it
68.6%
62.6%
62.2%
60.0%
58.6%
51.8%
51.8%
49.5%
49.0%
32.6%
30
40
50
60
70
drag reduction vs circle on 16 unseen seeds (%) · dots = runs, bar = mean (s.e. ≈ 0.1–0.2% per run)